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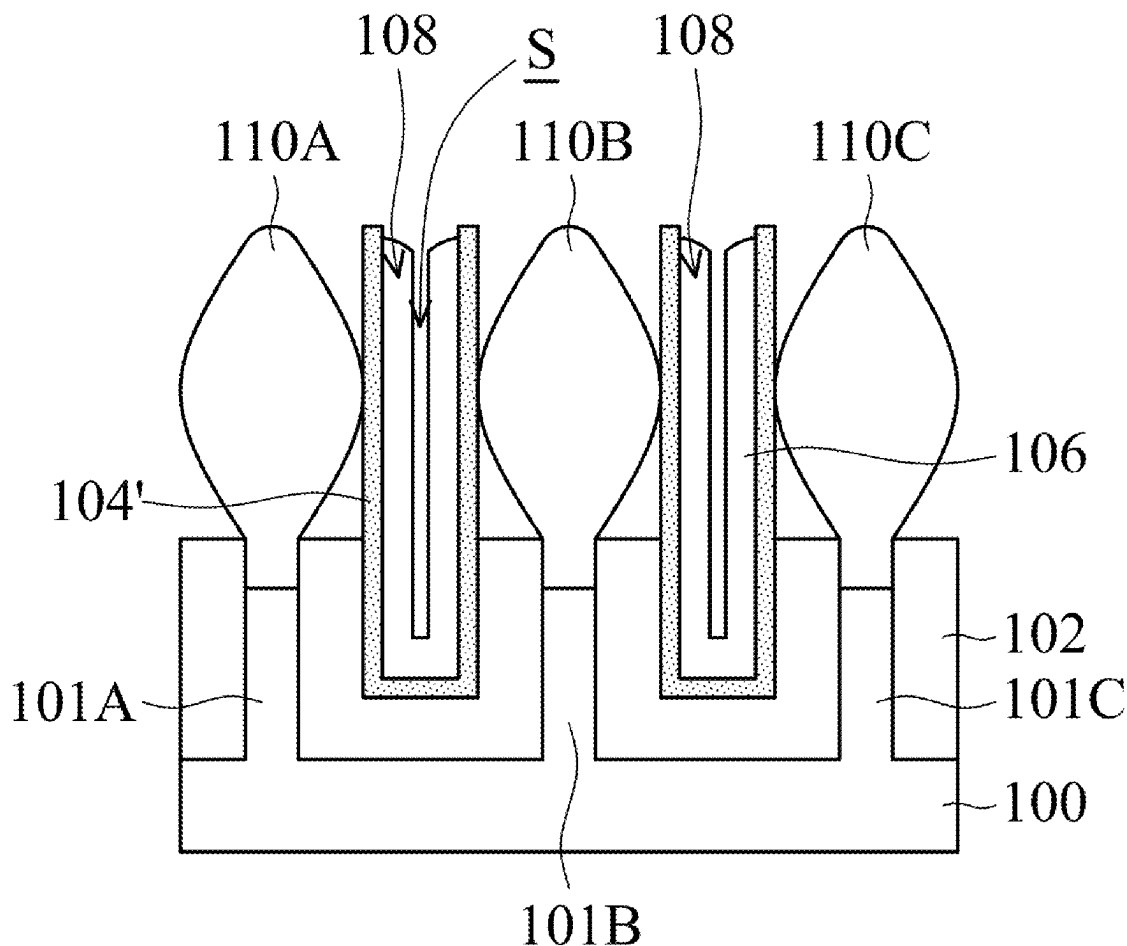
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84/0151 (2025.01); *H10D 84/834* (2025.01);
H01L 21/0214 (2013.01); *H01L 21/02148*
(2013.01); *H01L 21/02164* (2013.01); *H01L*
21/0217 (2013.01); *H01L 21/02181* (2013.01);
H01L 21/02183 (2013.01); *H01L 21/02189*
(2013.01); *H01L 21/02192* (2013.01); *H01L*
21/02194 (2013.01); *H01L 21/022* (2013.01);
H01L 21/02271 (2013.01)

- (22) Filed: **May 27, 2025**

(60) Continuation of application No. 18/318,195, filed on May 16, 2023, now Pat. No. 12,342,605, which is a division of application No. 17/350,282, filed on Jun. 17, 2021, now Pat. No. 11,688,645.

<i>H10D 84/03</i>	(2025.01)
<i>H01L 21/02</i>	(2006.01)
<i>H01L 21/762</i>	(2006.01)

A semiconductor device structure is provided. The semiconductor device structure includes a first semiconductor structure laterally spaced apart from a second semiconductor structure and a dielectric structure laterally between the first semiconductor structure and the second semiconductor structure. The dielectric structure has an inner portion and an outer layer extending along a sidewall of the inner portion. Average grain sizes of the outer layer and the inner portion are different. A topmost surface of the inner portion is vertically spaced apart from a topmost surface of the outer layer.



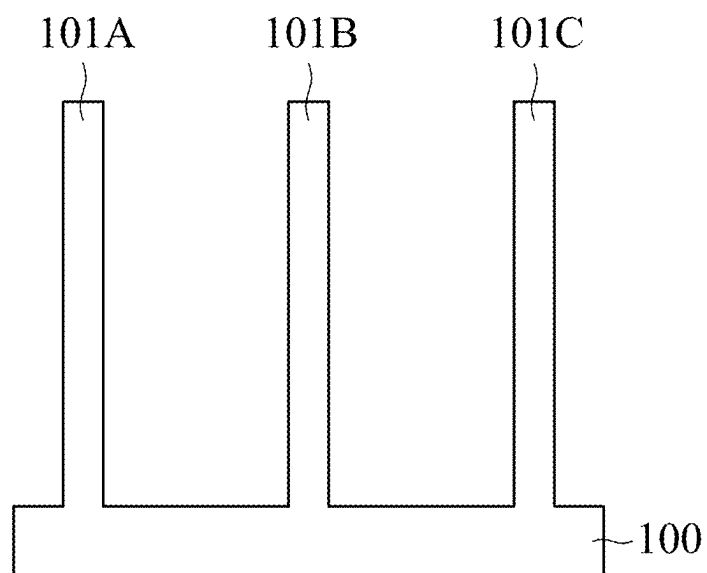


FIG. 1A

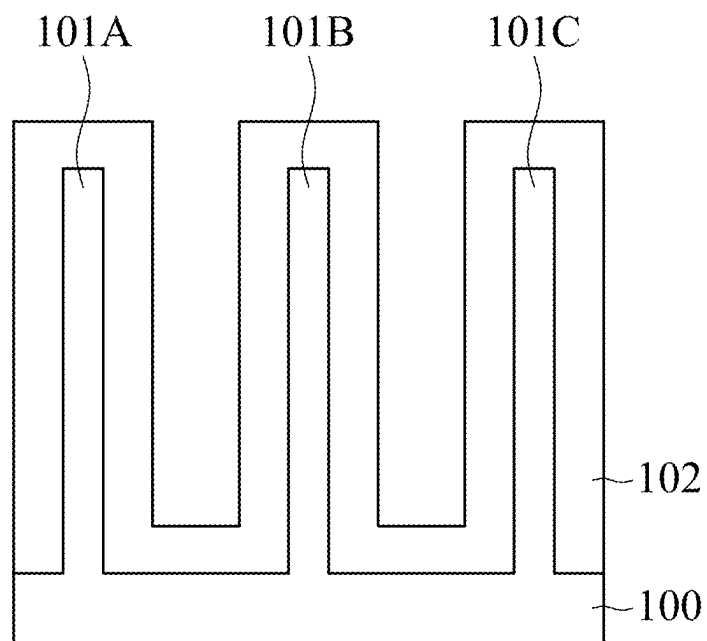


FIG. 1B

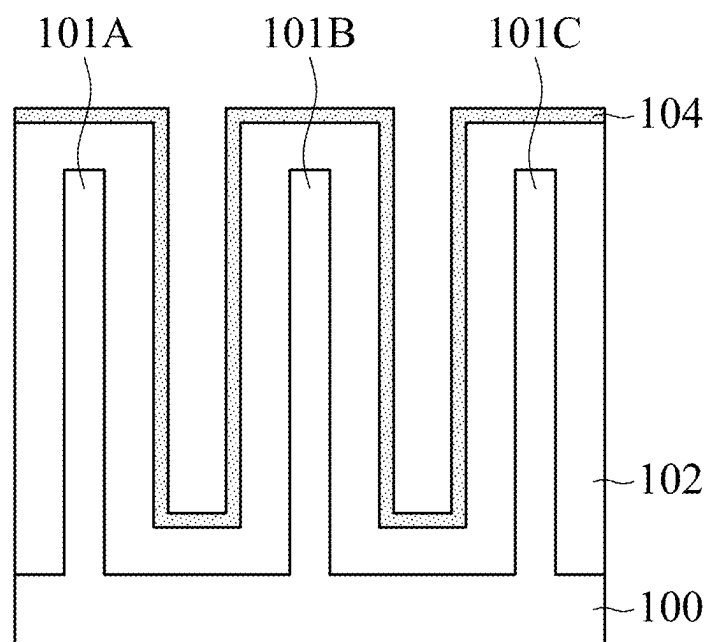


FIG. 1C

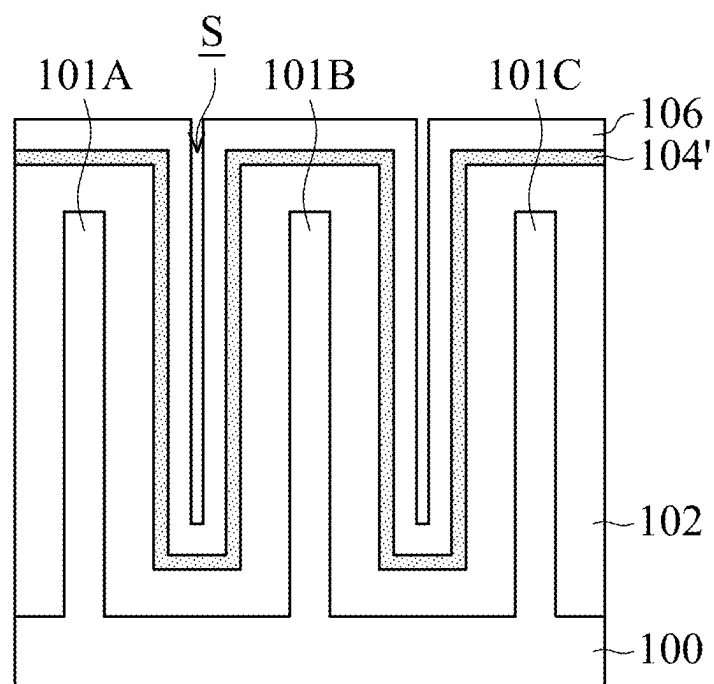


FIG. 1D

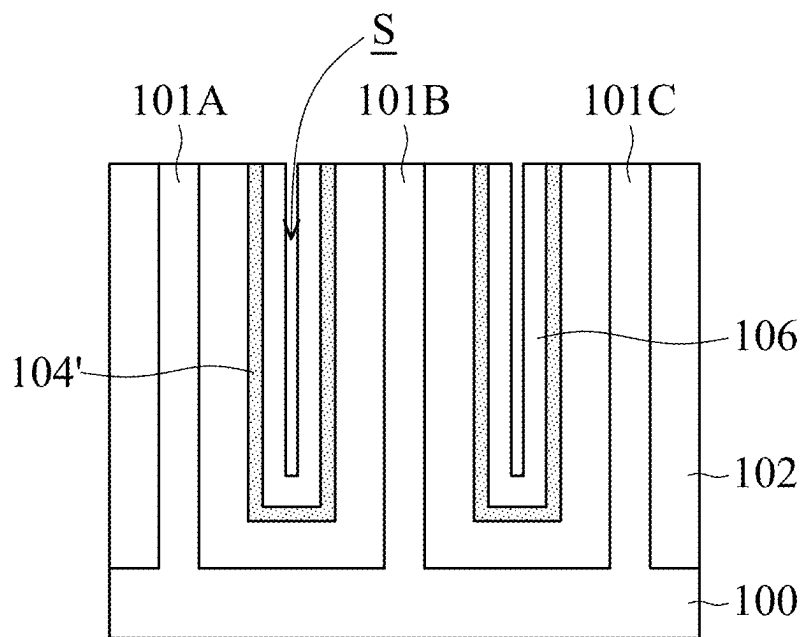


FIG. 1E

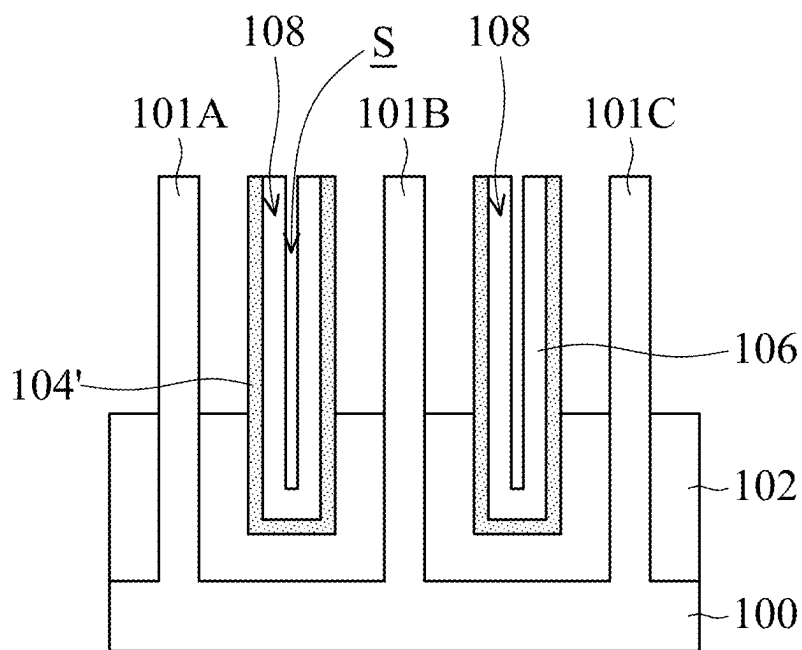


FIG. 1F

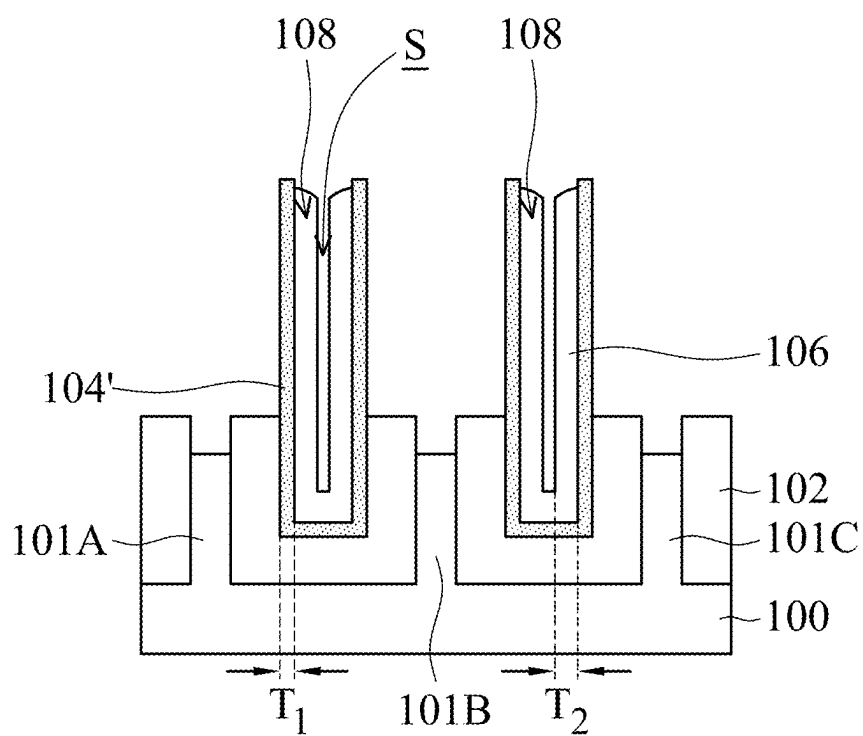


FIG. 1G

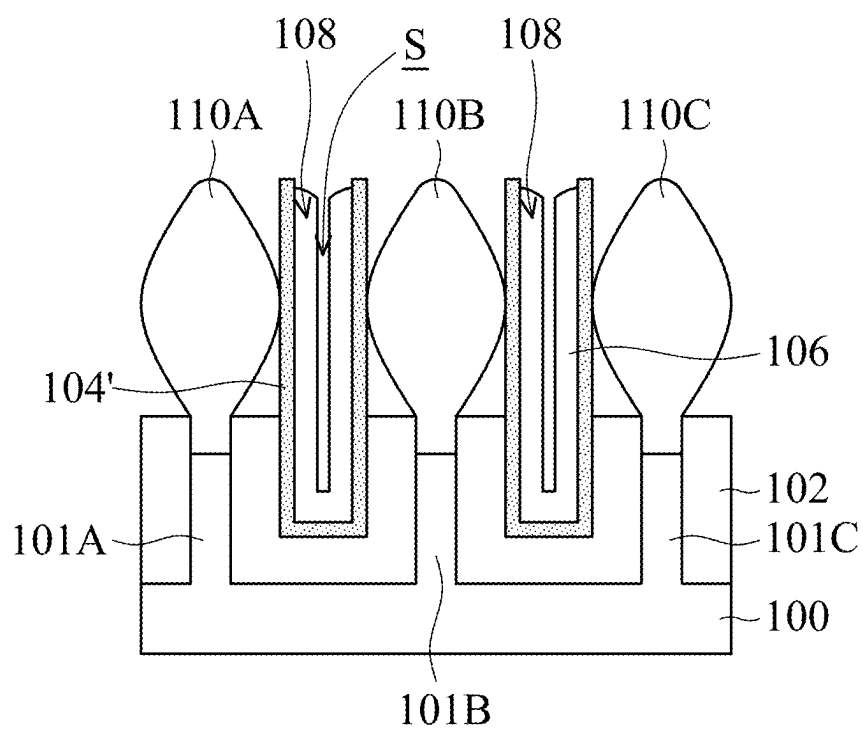


FIG. 1H

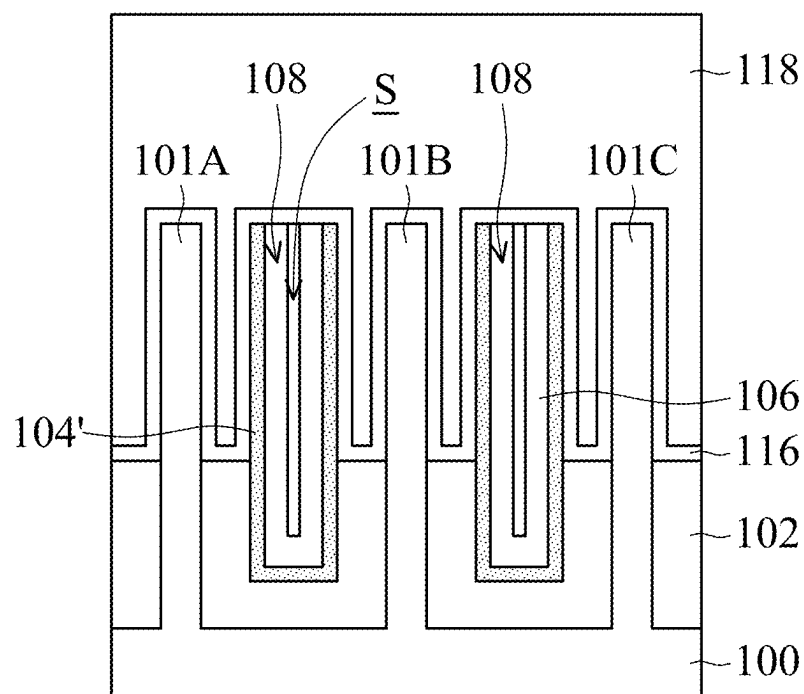


FIG. 2A

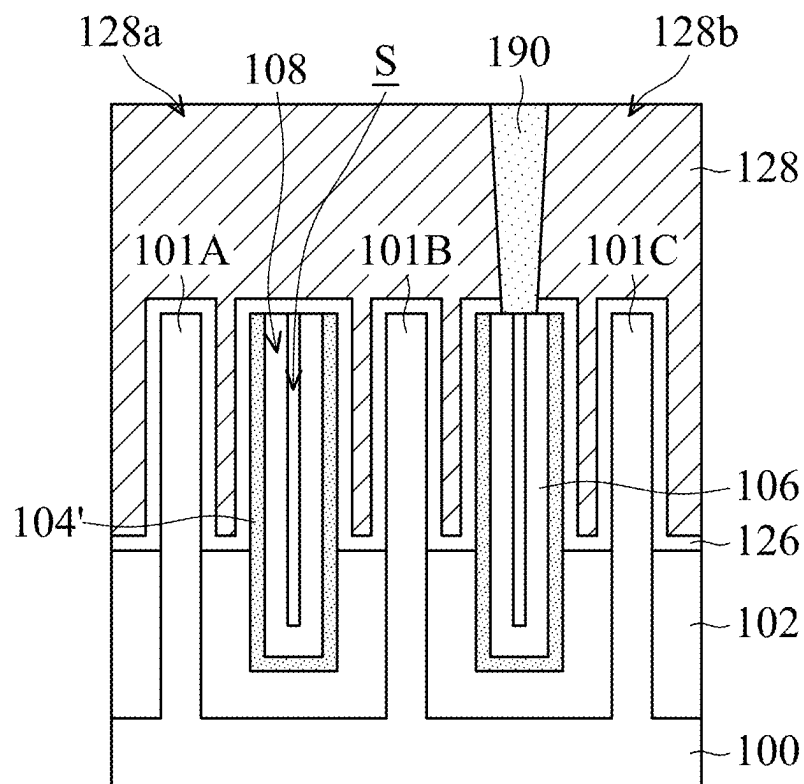


FIG. 2B

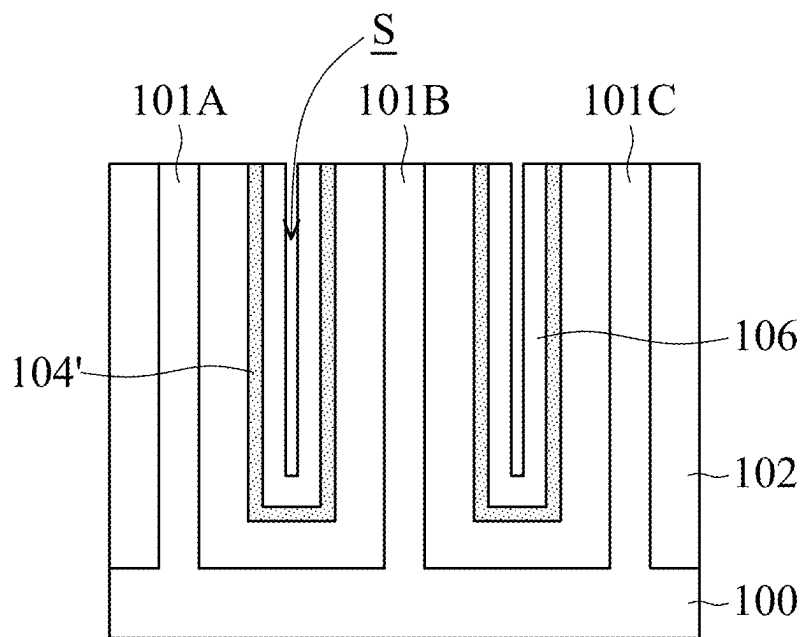


FIG. 3A

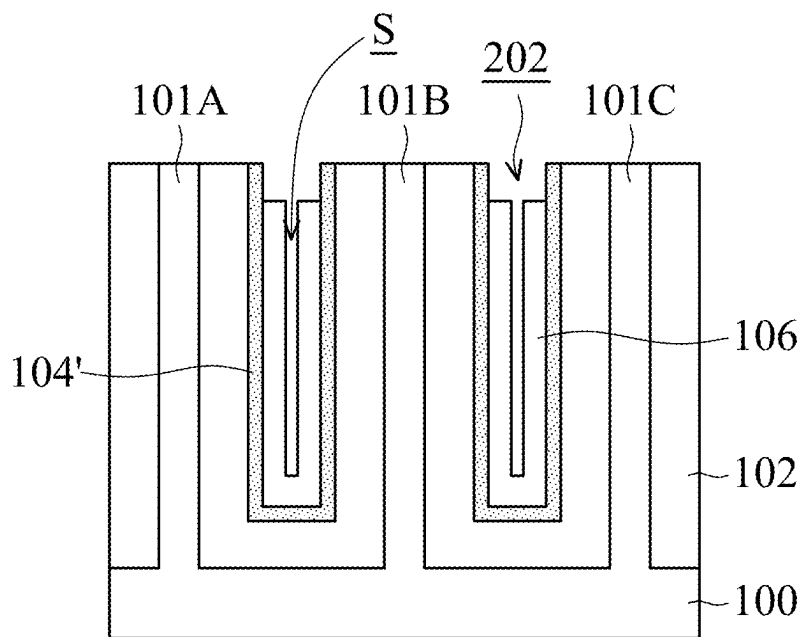


FIG. 3B

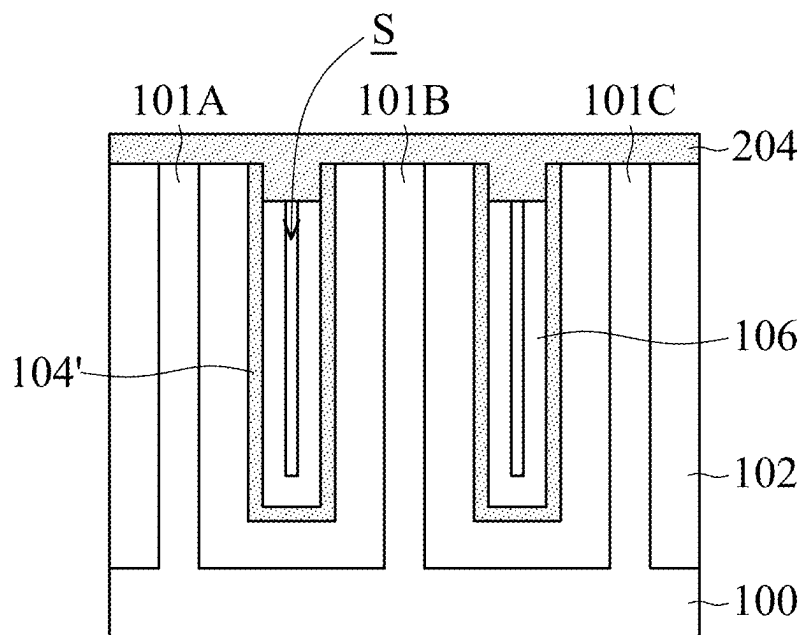


FIG. 3C

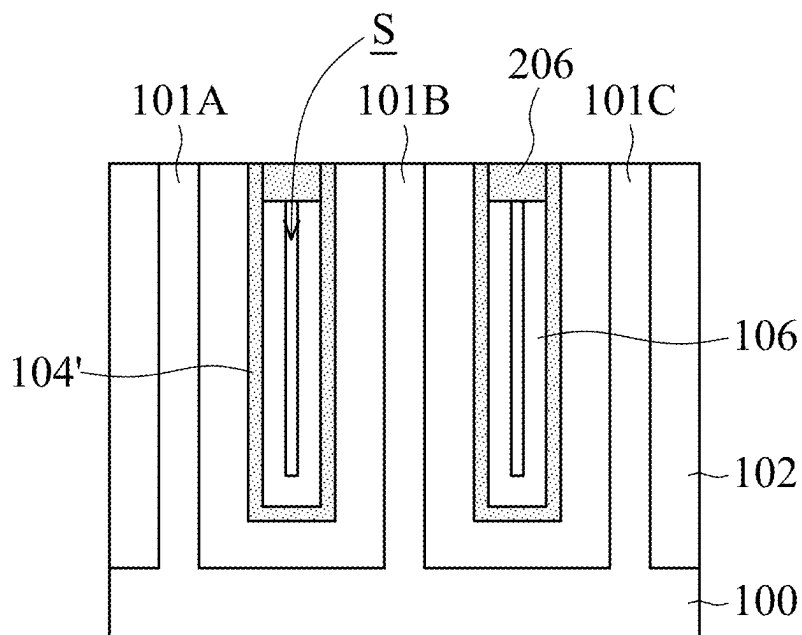


FIG. 3D

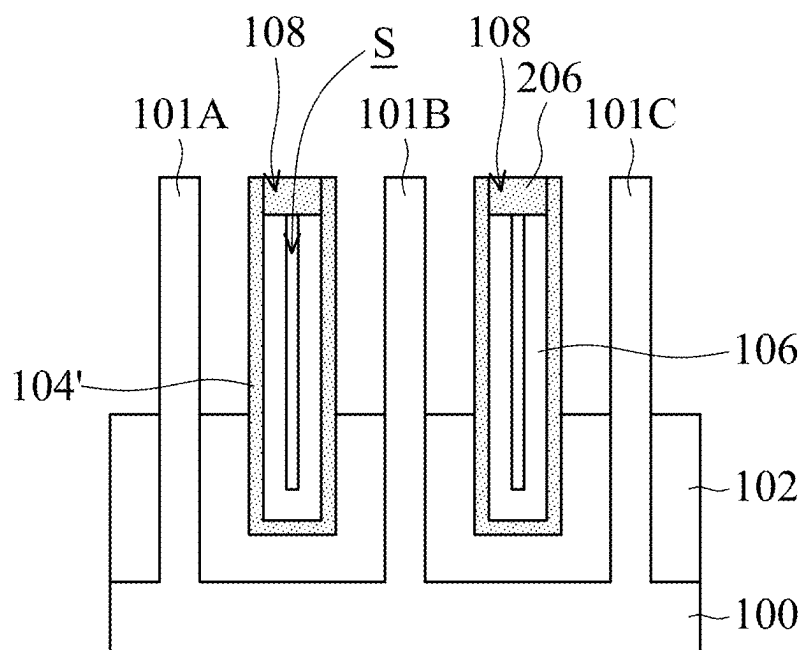


FIG. 3E

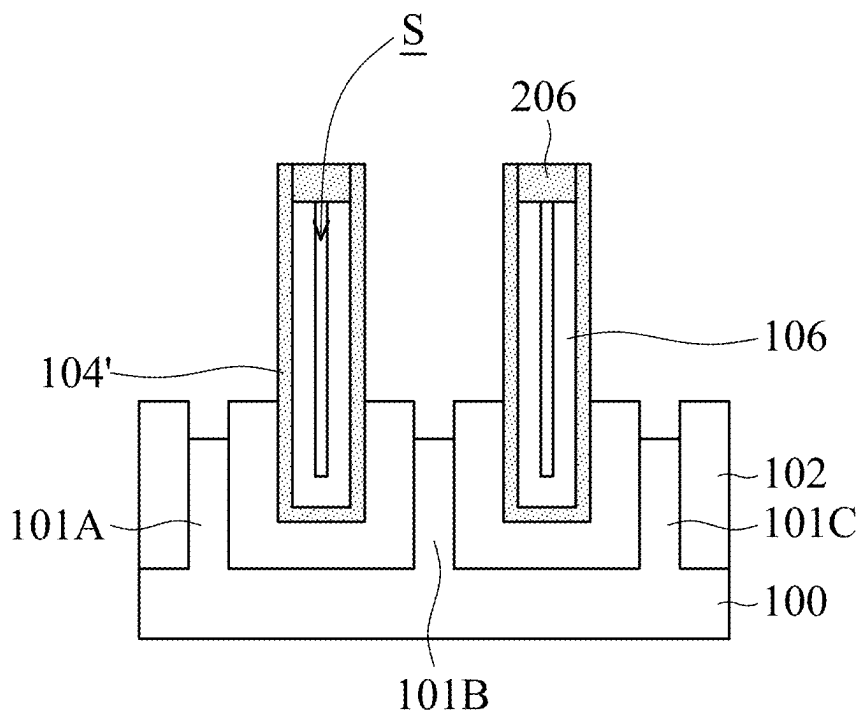


FIG. 3F

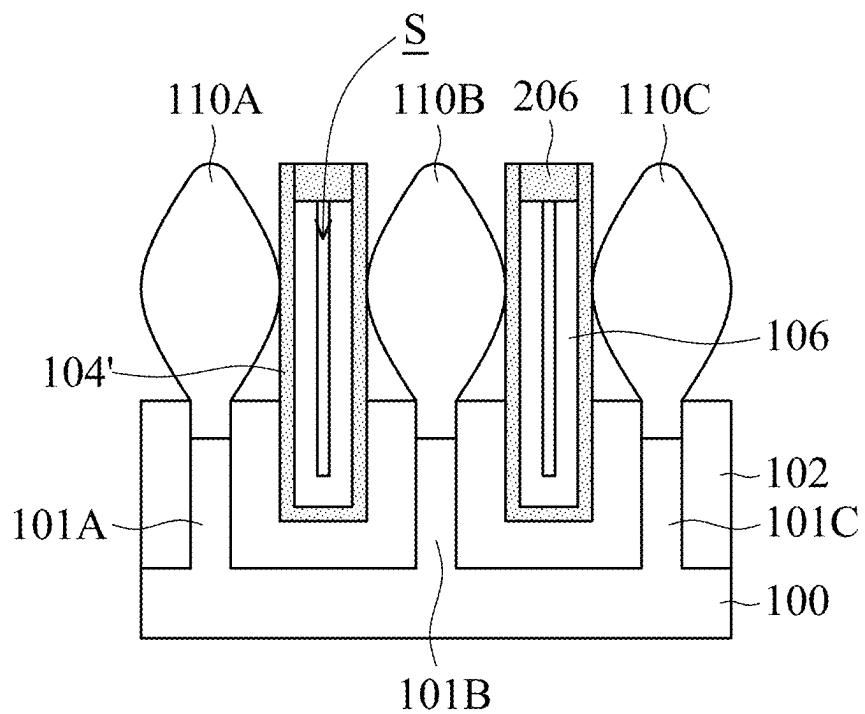


FIG. 3G

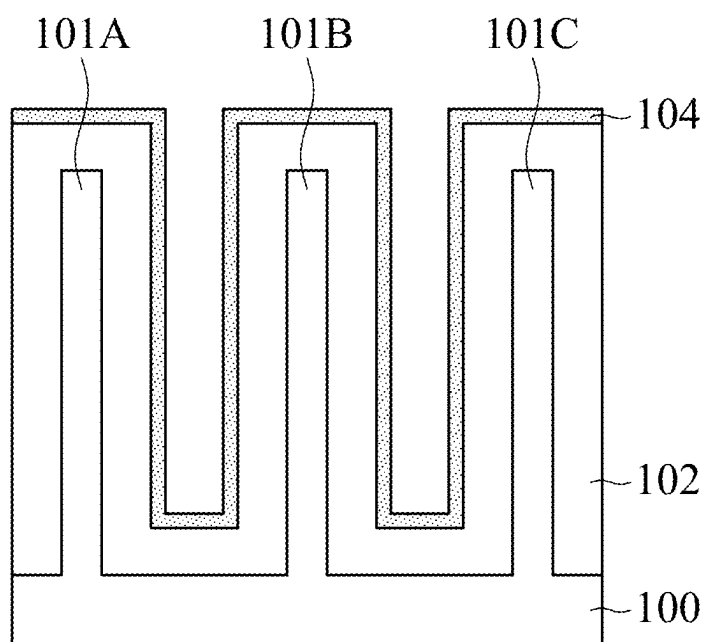


FIG. 4A

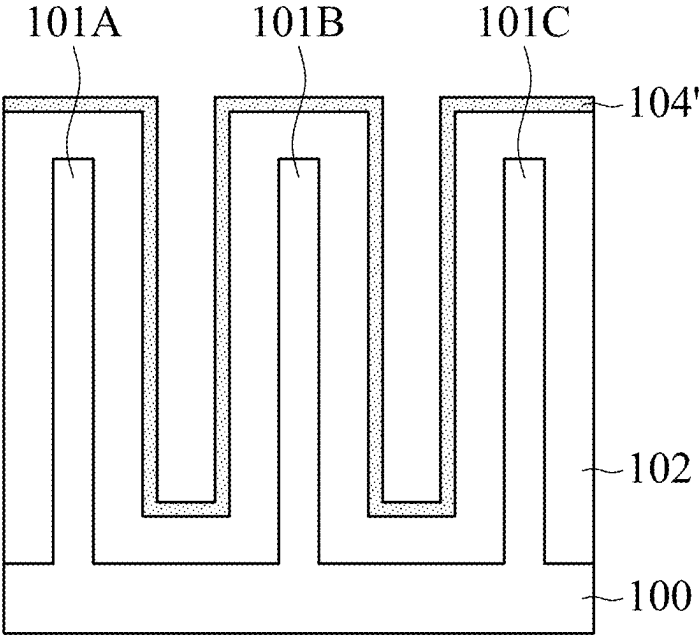


FIG. 4B

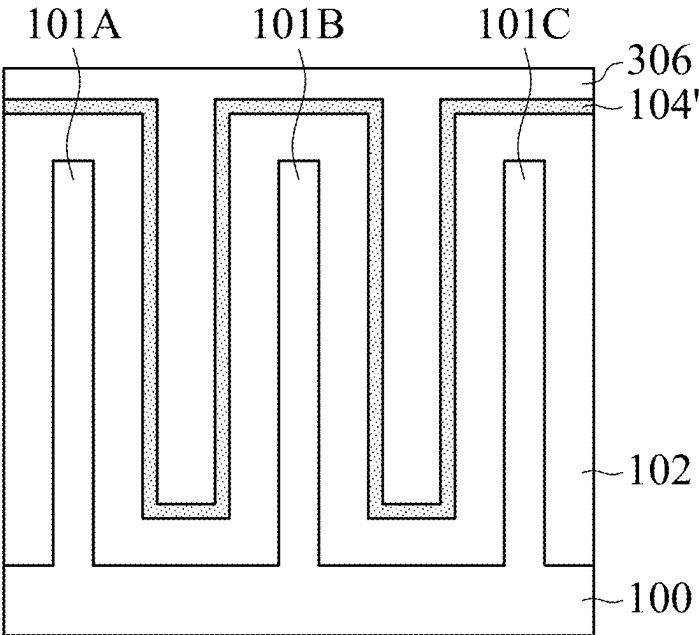


FIG. 4C

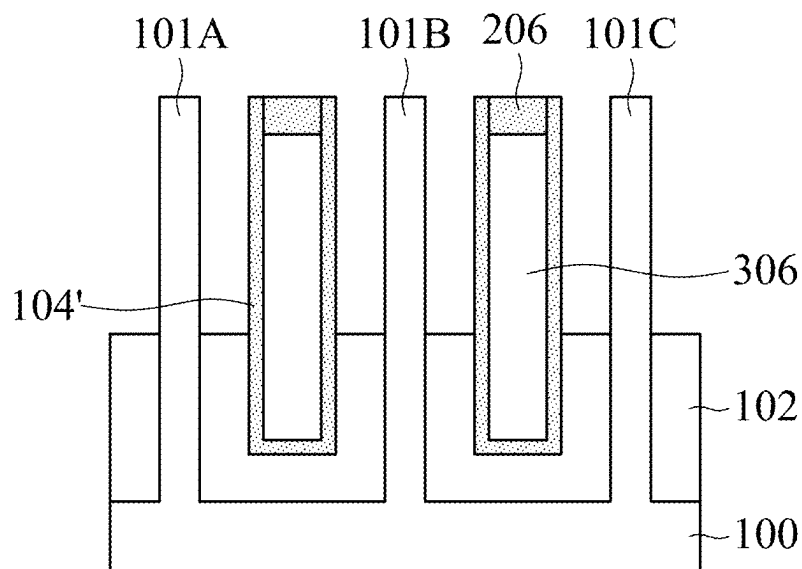


FIG. 4D

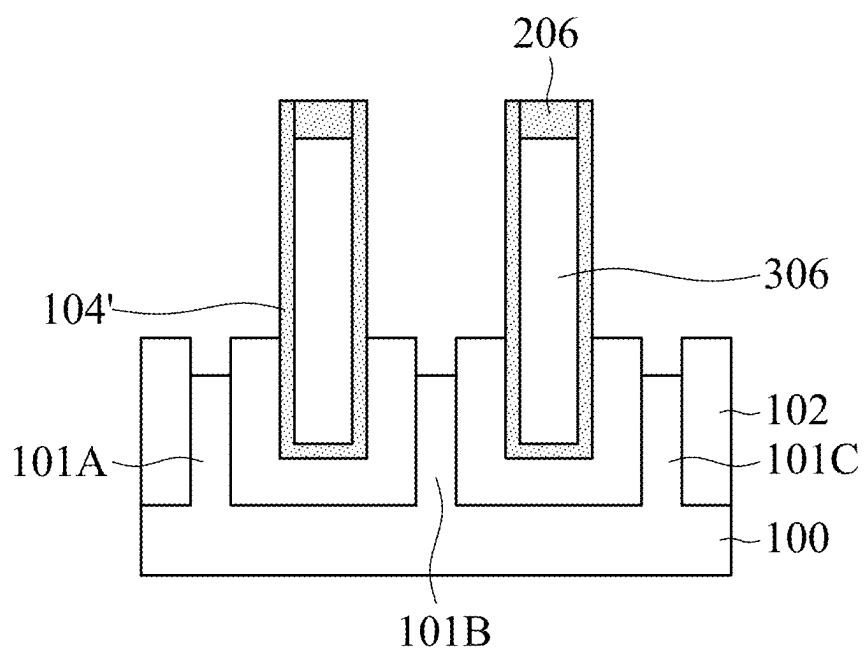


FIG. 4E

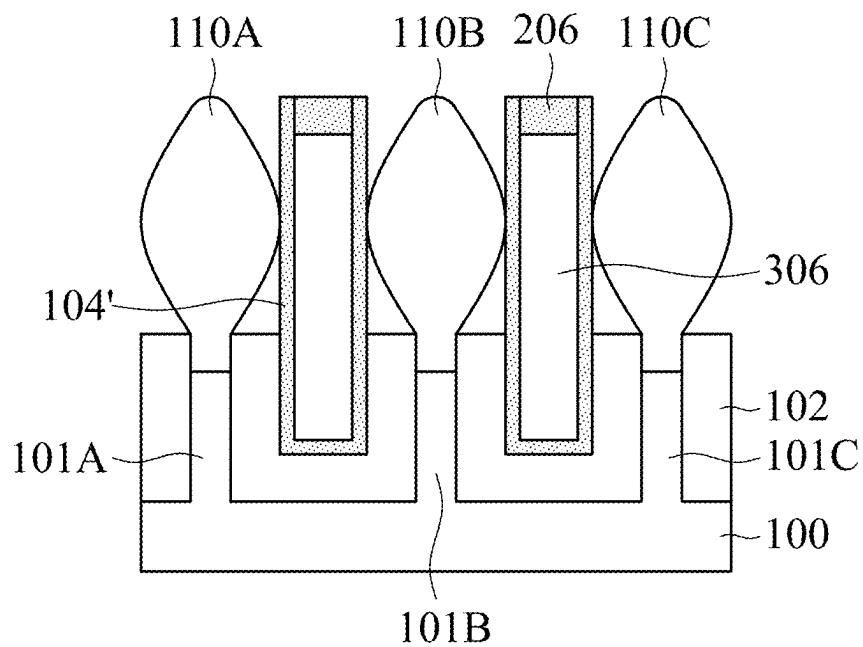


FIG. 4F

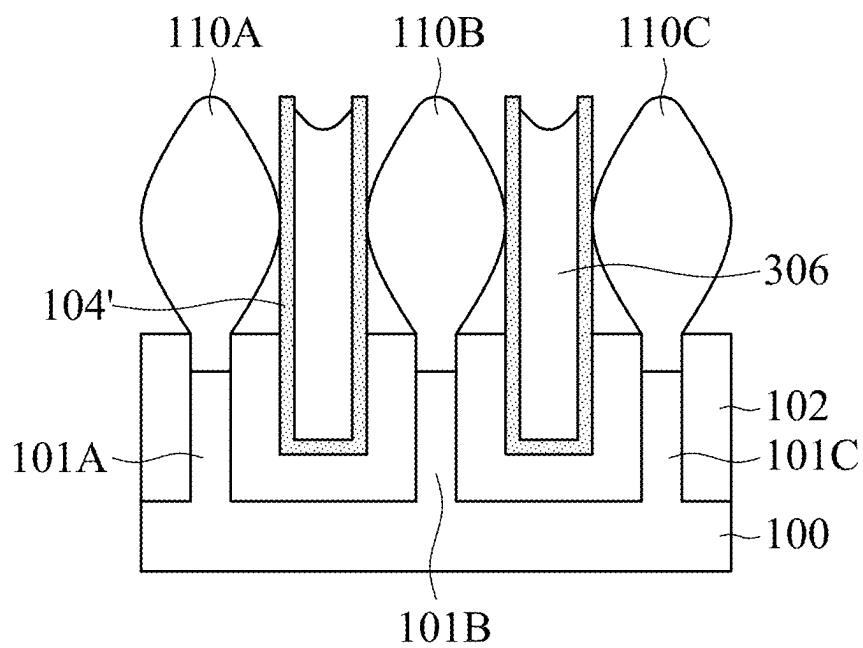


FIG. 5

SEMICONDUCTOR DEVICE WITH DIELECTRIC STRUCTURE

PRIORITY CLAIM AND CROSS-REFERENCE

[0001] This Application is a Continuation of U.S. application Ser. No. 18/318,195, filed on May 16, 2023, which is a Divisional of U.S. application Ser. No. 17/350,282, filed on Jun. 17, 2021 (now U.S. Pat. No. 11,688,645), the entirety of which are incorporated by reference herein.

BACKGROUND

[0002] The semiconductor integrated circuit (IC) industry has experienced rapid growth. Technological advances in IC materials and design have produced generations of ICs. Each generation has smaller and more complex circuits than the previous generation.

[0003] Over the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometric size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased. This scaling-down process generally provides benefits by increasing production efficiency and lowering associated costs.

[0004] However, these advances have increased the complexity of processing and manufacturing ICs. Since feature sizes continue to decrease, fabrication processes continue to become more difficult to perform. Therefore, it is a challenge to form reliable semiconductor devices at smaller and smaller sizes.

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It should be noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0006] FIGS. 1A-1H are cross-sectional views of various stages of a process for forming a portion of a semiconductor device structure, in accordance with some embodiments.

[0007] FIGS. 2A-2B are cross-sectional views of various stages of a process for forming a portion of a semiconductor device structure, in accordance with some embodiments.

[0008] FIGS. 3A-3G are cross-sectional views of various stages of a process for forming a portion of a semiconductor device structure, in accordance with some embodiments.

[0009] FIGS. 4A-4F are cross-sectional views of various stages of a process for forming a portion of a semiconductor device structure, in accordance with some embodiments.

[0010] FIG. 5 is a cross-sectional view of an intermediate stage of a process for forming a portion of a semiconductor device structure, in accordance with some embodiments.

DETAILED DESCRIPTION

[0011] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in

which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0012] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0013] The term “substantially” in the description, such as in “substantially flat” or in “substantially coplanar”, etc., will be understood by the person skilled in the art. In some embodiments the adjective substantially may be removed. Where applicable, the term “substantially” may also include embodiments with “entirely”, “completely”, “all”, etc. Where applicable, the term “substantially” may also relate to 90% or higher, such as 95% or higher, especially 99% or higher, including 100%. Furthermore, terms such as “substantially parallel” or “substantially perpendicular” are to be interpreted as not to exclude insignificant deviation from the specified arrangement and may include for example deviations of up to 10° in some embodiments. The word “substantially” does not exclude “completely” e.g. a composition which is “substantially free” from Y may be completely free from Y in some embodiments.

[0014] Terms such as “about” in conjunction with a specific distance or size are to be interpreted so as not to exclude insignificant deviation from the specified distance or size and may include for example deviations of up to 10% in some embodiments. The term “about” in relation to a numerical value x may mean $x \pm 5$ or 10% in some embodiments.

[0015] Terms such as “about” in conjunction with a specific distance or size are to be interpreted so as not to exclude insignificant deviation from the specified distance or size and may include for example deviations of up to 10%. The term “about” in relation to a numerical value x may mean $x \pm 5$ or 10%.

[0016] Embodiments of the disclosure may relate to Fin-FET structure having fins. The fins may be patterned using any suitable method. For example, the fins may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in some embodiments, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed,

and the remaining spacers may then be used to pattern the fins. However, the fins may be formed using one or more other applicable processes.

[0017] Some embodiments of the disclosure are described. Additional operations can be provided before, during, and/or after the stages described in these embodiments. Some of the stages that are described can be replaced or eliminated for different embodiments. Additional features can be added to the semiconductor device structure. Some of the features described below can be replaced or eliminated for different embodiments. Although some embodiments are discussed with operations performed in a particular order, these operations may be performed in another logical order.

[0018] FIGS. 1A-1H are cross-sectional views of various stages of a process for forming a portion of a semiconductor device structure, in accordance with some embodiments. As shown in FIG. 1A, a semiconductor substrate **100** is received or provided. In some embodiments, the semiconductor substrate **100** is a bulk semiconductor substrate, such as a semiconductor wafer.

[0019] The semiconductor substrate **100** may include silicon or other elementary semiconductor materials such as germanium. The semiconductor substrate **100** may be undoped or doped (e.g., p-type, n-type, or a combination thereof). In some embodiments, the semiconductor substrate **100** includes an epitaxially grown semiconductor layer on a dielectric layer. The epitaxially grown semiconductor layer may be made of silicon germanium, silicon, germanium, one or more other suitable materials, or a combination thereof.

[0020] In some other embodiments, the semiconductor substrate **100** includes a compound semiconductor. For example, the compound semiconductor includes one or more III-V compound semiconductors having a composition defined by the formula $Al_{X1}Ga_{X2}In_{X3}As_{Y1}P_{Y2}N_{Y3}Sb_{Y4}$, where X1, X2, X3, Y1, Y2, Y3, and Y4 represent relative proportions. Each of them is greater than or equal to zero, and added together they equal 1. The compound semiconductor may include silicon carbide, gallium arsenide, indium arsenide, indium phosphide, one or more other suitable compound semiconductors, or a combination thereof. Another suitable substrate including II-VI compound semiconductors may also be used.

[0021] In some embodiments, the semiconductor substrate **100** is an active layer of a semiconductor-on-insulator (SOI) substrate. The SOI substrate may be fabricated using a separation by implantation of oxygen (SIMOX) process, a wafer bonding process, another applicable method, or a combination thereof. In some other embodiments, the semiconductor substrate **100** includes a multi-layered structure. For example, the semiconductor substrate **100** includes a silicon-germanium layer formed on a bulk silicon layer.

[0022] As shown in FIG. 1A, multiple fin structures **101A**, **101B**, and **101C** are formed over the semiconductor substrate **100**, in accordance with some embodiments. In some embodiments, multiple recesses (or trenches) are formed in the semiconductor substrate **100**, in accordance with some embodiments. As a result, multiple fin structures that protrude from the surface of the semiconductor substrate **100** are formed or defined between the recesses. In some embodiments, one or more photolithography and etching processes are used to form the recesses. In some embodiments, the fin structures **101A** to **101C** are in direct contact with the semiconductor substrate **100**.

[0023] However, embodiments of the disclosure have many variations and/or modifications. In some other embodiments, the fin structures **101A** to **101C** are not in direct contact with the semiconductor substrate **100**. One or more other material layers may be formed between the semiconductor substrate **100** and the fin structures **101A** to **101C**. For example, a dielectric layer may be formed therebetween.

[0024] As shown in FIG. 1B, an isolation layer **102** is deposited over the semiconductor substrate **100** and the fin structures **101A** to **101C**, in accordance with some embodiments. In some embodiments, the isolation layer **102** extends along sidewalls and tops of the fin structures **101A** to **101C**. In some embodiments, the isolation layer **102** conformally extends along the fin structures **101A** to **101C**. In some embodiments, the isolation layer **102** is in direct contact with the fin structures **101A** to **101C**.

[0025] In some embodiments, the isolation layer **102** is made of or includes silicon oxide, silicon oxynitride, carbon-containing silicon oxide, one or more other suitable materials, or a combination thereof. The isolation layer **102** may be deposited using a chemical vapor deposition (CVD) process, an atomic layer deposition (ALD) process, a thermal oxidation process, one or more other applicable processes, or a combination thereof.

[0026] As shown in FIG. 1C, a protective layer **104** is deposited over the isolation layer **102**, in accordance with some embodiments. In some embodiments, the protective layer **104** extends conformally along the isolation layer **102**. In some embodiments, the protective layer **104** extends along sidewalls and tops of the fin structures **101A** to **101C**. In some embodiments, the protective layer **104** conformally extends along the fin structures **101A** to **101C**. In some embodiments, the protective layer **104** is in direct contact with the isolation layer **102**.

[0027] In some embodiments, the protective layer **104** is made of or includes a high dielectric constant (high-k) material. The high-k material may include hafnium oxide, hafnium zirconium oxide, zirconium oxide, hafnium dioxide-alumina alloy, hafnium silicon oxide, hafnium silicon oxynitride, hafnium tantalum oxide, hafnium titanium oxide, lanthanum oxide, hafnium lanthanum oxide, one or more other suitable materials, or a combination thereof. The protective layer **104** may be deposited using an ALD process, a CVD process, one or more other applicable processes, or a combination thereof.

[0028] As shown in FIG. 1D, a dielectric layer **106** is deposited over the protective layer **104**, in accordance with some embodiments. In some embodiments, the dielectric layer **106** extends conformally along the protective layer **104**. In some embodiments, the dielectric layer **106** extends along sidewalls and tops of the fin structures **101A** to **101C**. In some embodiments, the dielectric layer **106** conformally extends along the fin structures **101A** to **101C**. In some embodiments, the dielectric layer **106** is in direct contact with the protective layer **104**. In some embodiments, the dielectric layer **106** surrounds seams S, as shown in FIG. 1D.

[0029] In some embodiments, the dielectric layer **106** is made of or includes silicon nitride, silicon oxide, silicon oxynitride, carbon-containing silicon oxide, carbon-containing silicon oxynitride, one or more other suitable materials, or a combination thereof. The dielectric layer **106** may be deposited using an ALD process, a CVD process, one or more other applicable processes, or a combination thereof.

In some embodiments, the deposition of the dielectric layer 106 involves a thermal operation. For example, the deposition of the dielectric layer 106 is performed at a high temperature that is in a range from about 450 degrees C. to about 550 degrees C.

[0030] In some embodiments, due to the thermal operation, the protective layer 104 is crystallized during the deposition of the dielectric layer 106. As a result, the protective layer 104 is crystallized to form a crystallized protective layer 104', as shown in FIG. 1D in accordance with some embodiments. The crystallized protective layer 104' may have a better etching resistance than the protective layer 104 that is not or merely slightly crystallized. The crystallized protective layer 104' may thus provide better protection to dielectric fins that will be formed later.

[0031] In some embodiments, the dielectric layer 106 is substantially not crystallized. The dielectric layer 106 may be amorphous or include a small amount of nanocrystals. The crystallized protective layer 104' may have a first average grain size, and the dielectric layer 106 may have a second average grain size. In some embodiments, the first average grain size is larger than the second average grain size.

[0032] Many variations and/or modifications can be made to embodiments of the disclosure. In some embodiments, one or more additional thermal operations are used to form the crystallized protective layer 104' and/or to enhance the crystallization of the crystallized protective layer 104'. The operation temperature of the thermal operations may be in a range from about 450 degrees C. to about 1000 degrees C. The operation time of the thermal operations may be in a range from about 1 second to about 1 hour.

[0033] The crystallized protective layer 104' has a first dielectric constant, and the dielectric layer 106 has a second dielectric constant. In some embodiments, the first dielectric constant is higher than the second dielectric constant. In some embodiments, the first dielectric constant is higher than about 14. In some embodiments, the first dielectric constant is higher than the dielectric constant of silicon nitride. In some embodiments, the first dielectric constant is higher than the dielectric constant of carbon-containing silicon nitride. In some embodiments, the first dielectric constant is higher than the dielectric constant of carbon-containing silicon oxynitride.

[0034] As shown in FIG. 1E, the dielectric layer 106, the crystallized protective layer 104', and the isolation layer 102 are partially removed, in accordance with some embodiments. As a result, the tops of the fin structures 101A to 101C that are originally covered by these layers are exposed, as shown in FIG. 1E. In some embodiments, the dielectric layer 106, the crystallized protective layer 104', and the isolation layer 102 are partially removed using a planarization process. The planarization process may include a chemical mechanical polishing (CMP) process, a grinding process, a dry polishing process, an etching process, one or more other applicable processes, or a combination thereof.

[0035] As shown in FIG. 1F, the isolation layer 102 is recessed, in accordance with some embodiments. The isolation layer 102 may be recessed using an etching back process. As a result, the remaining portion of the isolation layer 102 forms an isolation structure. The isolation structure (i.e., the isolation layer 102) laterally surrounds lower portions of the fin structures 101A to 101C, as shown in FIG.

1F. The reference number "102" may also be used to designate the isolation structure.

[0036] As shown in FIG. 1F, the remaining portions of the crystallized protective layer 104' and the dielectric layer 106 together form multiple dielectric fins 108. The isolation structure (i.e., the isolation layer 102) also laterally surrounds lower portions of the dielectric fins 108. Each of the fin structures 101A to 101C and each of the dielectric fins 108 protrude from the top surface of the isolation structure (i.e., the isolation layer 102). In some embodiments, in each of the dielectric fins 108, the seam S extends downwards from the top surface of the inner portion (i.e., the dielectric layer 106) to exceed the topmost surface of the isolation structure 102, as shown in FIG. 1F.

[0037] The crystallized protective layer 104' of the dielectric fin 108 may function as a protective shell that protects the inner portion of the dielectric fin 108. The inner portion of the dielectric fin 108 is a dielectric structure constructed by the dielectric layer 106. During a subsequent etching process, the crystallized protective layer 104' that has good etching resistance may protect the dielectric layer 106 from being damaged. Due to the blocking of the dielectric layer 106, voids are prevented from being formed in the crystallized protective layer 104'.

[0038] In some other cases where the dielectric layer 106 is not formed, the crystallization of the protective layer 104 may occur in accompany with grain growth. As a result, the seam S may be randomly merged into voids. These voids may result in the unexpected merging of nearby epitaxial structures. Alternatively, the voids may result in unexpected short circuiting between two portions of a gate stack that are designed to be electrically isolated from each other.

[0039] FIGS. 2A-2B are cross-sectional views of various stages of a process for forming a portion of a semiconductor device structure, in accordance with some embodiments. As shown in FIG. 2A, after the formation of the dielectric fins 108, a dummy gate stack is formed over the fin structures 101A to 101C and the dielectric fins 108, in accordance with some embodiments. The dummy gate stack is formed to partially cover and to extend across the fin structures 101A to 101C.

[0040] The dummy gate stack includes a dummy gate dielectric layer 116 and a dummy gate electrode 118. The dummy gate dielectric layer 116 may be made of or include silicon oxide. The dummy gate electrode 118 may be made of or include polysilicon. In some embodiments, a dummy gate dielectric material layer and a dummy gate electrode layer are sequentially deposited over the isolation structure 102, the fin structures 101A to 101C, and the dielectric fins 108. Afterwards, the dummy gate dielectric material layer and the dummy gate electrode layer are patterned to form the dummy gate stack.

[0041] FIG. 1G shows the cross-sectional view of the portion of the semiconductor device structure that is not covered by the dummy gate stack. The cross-sectional view is taken along a line parallel to the longitudinal direction of the dummy gate stack but not overlapping the dummy gate stack.

[0042] As shown in FIG. 1G, after the formation of the dummy gate stack as illustrated in FIG. 2A, the exposed portions of the fin structures 101A to 101C are partially removed, in accordance with some embodiments. The fin structures 101A to 101C may be recessed using one or more etching processes. In some embodiments, after the recess-

ing, the top surfaces of the fin structures **101A** to **101C** are lower than the top surface of the isolation structure **102**.

[0043] In some embodiments, the dielectric layer **106** of the dielectric fin **108** is slightly etched during the recessing of the fin structures **101A** to **101C**. In some embodiments, the topmost surface of the inner portion (i.e., the dielectric layer **106**) of the dielectric fin **108** is closer to the semiconductor substrate **100** than the topmost surface of the crystallized protective layer **104'** (that functions as a protective shell) of the dielectric fin **108**. In some embodiments, the topmost surface of the inner portion (i.e., the dielectric layer **106**) has a curved profile.

[0044] As shown in FIG. 1G, the crystallized protective layer **104'** has a film thickness T_1 , and the dielectric layer **106** has a film thickness T_2 . In some embodiments, the film thickness T_1 of the crystallized protective layer **104'** is in a range from about 2 nm to about 5 nm. In some embodiments, the film thickness T_2 of the dielectric layer **106** is in a range from about 2 nm to about 10 nm. The ratio (T_1/T_2) of the film thickness T_1 to the film thickness T_2 may be in a range from about 0.5 to about 1. In some embodiments, the ratio of the film thickness T_1 to the total width of the dielectric fin **108** may be in a range from about 0.2 to about 0.4.

[0045] In some cases where the film thickness T_1 of the crystallized protective layer **104'** is thinner than about 2 nm, the grain growth of the crystallized protective layer **104'** might be suppressed. The crystallized protective layer **104'** might not be able to provide sufficient etch resistance to sustain the subsequent etching process. In some other cases where the film thickness T_1 of the crystallized protective layer **104'** is thicker than about 5 nm, voids might be formed in the crystallized protective layer **104'** since the seam would be too narrow. As a result, the seam **S** may be randomly merged into voids. As mentioned above, these voids may result in the unexpected merging of nearby epitaxial structures. Alternatively, the voids may result in unexpected short circuiting between two portions of a gate stack that are designed to be electrically isolated from each other.

[0046] As shown in FIG. 1H, epitaxial structures **110A**, **110B**, and **110C** are respectively formed on the recessed fin structures **101A**, **110B**, and **110C**, in accordance with some embodiments. Due to the dielectric fins **108**, each of the epitaxial structures **110A**, **110B**, and **110C** is prevented from being merged with another nearby epitaxial structure. In some embodiments, the dielectric fins **108** are in direct contact with the nearby epitaxial structures **110A**, **110B**, and/or **110C**.

[0047] In some embodiments, the epitaxial structures **110A** to **110C** are p-type semiconductor structures. For example, the epitaxial structures **110A** to **110C** may include epitaxially grown silicon germanium or silicon germanium doped with boron. It should be appreciated, however, that the epitaxial structures **110A** to **110C** are not limited to being p-type semiconductor structures.

[0048] In some embodiments, the epitaxial structures **110A** to **110C** are n-type semiconductor structures. The epitaxial structures **110A** to **110C** may include epitaxially grown silicon, epitaxially grown silicon carbide (SiC), epitaxially grown silicon phosphide (SiP), or another suitable epitaxially grown semiconductor material. Alternatively, one or two of the epitaxial structures **110A** to **110C** is a p-type semiconductor structure while another one is an n-type semiconductor structure.

[0049] In some embodiments, the epitaxial structures **110A** to **110C** are formed by using a selective epitaxial growth (SEG) process, a CVD process (e.g., a vapor-phase epitaxy (VPE) process, a low pressure chemical vapor deposition (LPCVD) process, and/or an ultra-high vacuum CVD (UHV-CVD) process), a molecular beam epitaxy process, one or more other applicable processes, or a combination thereof.

[0050] In some embodiments, one or both of the epitaxial structures **110A** to **110C** are doped with one or more suitable dopants. For example, the epitaxial structures **110A** to **110C** are SiGe source/drain features doped with boron (B), indium (In), or another suitable dopant. Alternatively, in some other embodiments, one or both of the epitaxial structures **110A** to **110C** are Si source/drain features doped with phosphor (P), antimony (Sb), or another suitable dopant.

[0051] In some embodiments, the epitaxial structures **110A** to **110C** are doped in-situ during their epitaxial growth. In some other embodiments, the epitaxial structures **110A** to **110C** are not doped during the growth of the epitaxial structures **110A** to **110C**. Instead, after the formation of the epitaxial structures **110A** to **110C**, the epitaxial structures **110A** to **110C** are doped in a subsequent process. In some embodiments, the doping is achieved by using an ion implantation process, a plasma immersion ion implantation process, a gas and/or solid source diffusion process, one or more other applicable processes, or a combination thereof. In some embodiments, the epitaxial structures **110A** to **110C** are further exposed to one or more annealing processes to activate the dopants. For example, a rapid thermal annealing process is used.

[0052] After the formation of the epitaxial structures **110A** to **110C**, a dielectric layer is deposited over the dummy gate stack and the epitaxial structures **110A** to **110C**. Afterwards, a planarization process is used to thin down the dielectric layer and to expose the top surface of the dummy gate stack. Then, the dummy gate stack is removed to form a trench that partially exposes the fin structures **101A** to **101C** and the dielectric fins **108**. Afterwards, a metal gate stack is formed in the trench.

[0053] As shown in FIG. 2B, the metal gate stack includes a gate dielectric layer **126** and a metal gate electrode **128**. The metal gate electrode **128** may include a work function layer and a conductive filling. In some embodiments, the formation of the metal gate stack involves the deposition of multiple metal gate stack layers over the dielectric layer to fill the trench.

[0054] In some embodiments, the gate dielectric layer **126** is made of or includes a dielectric material with high dielectric constant (high-K). The gate dielectric layer **126** may be made of or include hafnium oxide, zirconium oxide, aluminum oxide, hafnium dioxide-alumina alloy, hafnium silicon oxide, hafnium silicon oxynitride, hafnium tantalum oxide, hafnium titanium oxide, hafnium zirconium oxide, one or more other suitable high-K materials, or a combination thereof. The gate dielectric layer **126** may be deposited using an ALD process, a CVD process, one or more other applicable processes, or a combination thereof.

[0055] In some embodiments, before the formation of the gate dielectric layer **126**, an interfacial layer is formed on the surfaces of the fin structures **101A** to **101C**. The interfacial layer is very thin and is made of, for example, silicon oxide or germanium oxide. In some embodiments, the interfacial layer is formed by applying an oxidizing agent on the

surfaces of the fin structures **101A** to **101C**. For example, a hydrogen peroxide-containing liquid may be applied or provided on the surfaces of the fin structures **101A** to **101C** so as to grow the interfacial layer.

[0056] The work function layer may be used to provide the desired work function for transistors to enhance device performance including improved threshold voltage. In some embodiments, the work function layer is used for forming an NMOS device. The work function layer is an n-type work function layer. The n-type work function layer is capable of providing a work function value suitable for the device, such as equal to or less than about 4.5 eV.

[0057] The n-type work function layer may include metal, metal carbide, metal nitride, or a combination thereof. For example, the n-type work function layer includes titanium nitride, tantalum, tantalum nitride, one or more other suitable materials, or a combination thereof. In some embodiments, the n-type work function is an aluminum-containing layer. The aluminum-containing layer may be made of or include TiAlC, TiAlO, TiAlN, one or more other suitable materials, or a combination thereof.

[0058] The work function layer may also be made of or include hafnium, zirconium, titanium, tantalum, aluminum, metal carbides (e.g., hafnium carbide, zirconium carbide, titanium carbide, aluminum carbide), aluminides, ruthenium, palladium, platinum, cobalt, nickel, conductive metal oxides, or a combinations thereof. The thickness and/or the compositions of the work function layer may be fine-tuned to adjust the work function level.

[0059] The work function layer may be deposited over the gate dielectric layer **126** using an ALD process, a CVD process, a PVD process, an electroplating process, an electroless plating process, one or more other applicable processes, or a combination thereof.

[0060] In some embodiments, a barrier layer is formed before the work function layer to interface the gate dielectric layer **126** with the subsequently formed work function layer. The barrier layer may also be used to prevent diffusion between the gate dielectric layer **126** and the subsequently formed work function layer. The barrier layer may be made of or include a metal-containing material. The metal-containing material may include titanium nitride, tantalum nitride, one or more other suitable materials, or a combination thereof. The barrier layer may be deposited using an ALD process, a CVD process, a PVD process, an electroplating process, an electroless plating process, one or more other applicable processes, or a combination thereof.

[0061] In some embodiments, the conductive filling is made of or includes a metal material. The metal material may include tungsten, aluminum, copper, cobalt, one or more other suitable materials, or a combination thereof. A conductive layer used for forming the conductive filling may be deposited over the work function layer using a CVD process, an ALD process, a PVD process, an electroplating process, an electroless plating process, a spin coating process, one or more other applicable processes, or a combination thereof.

[0062] In some embodiments, a blocking layer is formed over the work function layer before the formation of the conductive layer used for forming the conductive filling. The blocking layer may be used to prevent the subsequently formed conductive layer from diffusing or penetrating into the work function layer. The blocking layer may be made of or include tantalum nitride, titanium nitride, one or more

other suitable materials, or a combination thereof. The blocking layer may be deposited using an ALD process, a PVD process, an electroplating process, an electroless plating process, one or more other applicable processes, or a combination thereof.

[0063] Afterwards, a planarization process is performed to remove the portions of the metal gate stack layers that are outside of the trench, in accordance with some embodiments. As a result, the remaining portions of the metal gate stack layers form the metal gate stack, as shown in FIG. 2B.

[0064] Afterwards, an insulating structure **190** is formed in the metal gate stack, as shown in FIG. 2B in accordance with some embodiments. In some embodiments, the insulating structure **190** penetrates through the metal gate electrode **128** and the gate dielectric layer **126** to reach one of the dielectric fins **108**. In some embodiments, the insulating structure **190** and the dielectric fin **108** thereunder together separate the metal gate stack into a first portion **128a** and a second portion **128b**, as shown in FIG. 2B. The first portion **128a** and the second portion **128b** are thus electrically isolated from each other. The first portion **128a** and the second portion **128b** that are cut from the same metal gate stack may now function as two metal gate stacks of different devices.

[0065] The insulating structure **190** may be made of or include silicon oxide, silicon nitride, silicon oxynitride, silicon carbide, carbon-containing silicon oxide, carbon-containing silicon nitride, carbon-containing silicon oxynitride, one or more other suitable materials, or a combination thereof. In some embodiments, a photolithography process and an etching process is used to partially remove the metal gate stack and to form a trench that exposes one of the dielectric fins **108**. Afterwards, the insulating material used for forming the insulating structure **190** is formed to fill the trench. A planarization process may then be used to remove the portion of the insulating material outside of the trench. As a result, the remaining portion of the insulating material in the trench forms the insulating structure **190**.

[0066] Many variations and/or modifications can be made to embodiments of the disclosure. FIGS. 3A-3G are cross-sectional views of various stages of a process for forming a portion of a semiconductor device structure, in accordance with some embodiments. As shown in FIG. 3A, a structure the same as or similar to the structure shown in FIG. 1E is formed, in accordance with some embodiments.

[0067] As shown in FIG. 3B, before the recessing of the isolation layer **102**, the dielectric layer **106** is partially removed to form recesses **202**, in accordance with some embodiments. In some embodiments, the recesses **202** connect the seam S.

[0068] As shown in FIG. 3C, a protective material layer **204** is deposited to fill the recesses **202**, in accordance with some embodiments. In some embodiments, the protective material layer **204** is made of or includes a high dielectric constant (high-k) material. The high-k material may include hafnium oxide, hafnium zirconium oxide, zirconium oxide, aluminum oxide, hafnium dioxide-alumina alloy, hafnium silicon oxide, hafnium silicon oxynitride, hafnium tantalum oxide, hafnium titanium oxide, one or more other suitable materials, or a combination thereof. The protective material layer **204** may be deposited using an ALD process, a CVD process, a flowable chemical vapor deposition (FCVD) process, one or more other applicable processes, or a combination thereof. The material and/or formation method of

the protective material layer 204 may be the same as or similar to those of the protective layer 104.

[0069] As shown in FIG. 3D, the portion of the protective material layer 204 outside of the recesses 202 is removed, in accordance with some embodiments. As a result, the remaining portions of the protective material layer 204 form protective caps 206 over the inner portions of the dielectric fins 108. In some embodiments, the seams S are sealed by the protective caps 206 and become closed holes. In some embodiments, the topmost surfaces of the protective caps 206 and the protective shells (i.e., the crystallized protective layer 104') are at substantially the same level, as shown in FIG. 3D. In some embodiments, the protective shell (i.e., the crystallized protective layer 104') extends along or covers the sidewalls of the protective cap 206, as shown in FIG. 3D.

[0070] As shown in FIG. 3E, similar to the process illustrated in FIG. 1F, the isolation layer 102 is partially removed, in accordance with some embodiments. As a result, the fin structures 101A to 101C and the dielectric fins 108 protrude from the top surface of the isolation structure (i.e., the remaining portion of the isolation layer 102).

[0071] Afterwards, similar to the embodiments illustrated in FIG. 2A, a dummy gate stack is formed to partially cover the fin structures 101A to 101B and the dielectric fins 108.

[0072] As shown in FIG. 3F, similar to the process illustrated in FIG. 1G, the portions of the fin structures 101A to 101C that are not covered by the dummy gate stack is then partially removed, in accordance with some embodiments. During the recessing of the fin structures 101A to 101C, the dielectric structure (i.e., the dielectric layer 106) of the dielectric fin 108 under the protective cap 206 is protected and prevented from being damaged by the etchant used for recessing the fin structures 101A to 101C.

[0073] In some embodiments, similar to the protective shell of the dielectric fin 108, one or more thermal operations is used to crystallize the protective cap 206, so as to enhance the etching resistance of the protective cap 206. In some embodiments, the thermal operation is performed after the deposition of the protective material layer 204. In some other embodiments, the thermal operation is performed after the formation of the protective cap 206 and before the recessing of the fin structures 101A to 101C.

[0074] As shown in FIG. 3G, similar to the embodiments illustrated in FIG. 1H, the epitaxial structures 110A to 110C are formed, in accordance with some embodiments. Due to the dielectric fins 108, the epitaxial structures 110A to 110C are prevented from being merged together. The performance and reliability of the semiconductor device structure are ensured.

[0075] Many variations and/or modifications can be made to embodiments of the disclosure. FIGS. 4A-4F are cross-sectional views of various stages of a process for forming a portion of a semiconductor device structure, in accordance with some embodiments. As shown in FIG. 4A, a structure the same as or similar to the structure shown in FIG. 1C is formed, in accordance with some embodiments.

[0076] As shown in FIG. 4B, a thermal operation is used to crystallize the protective layer 104, so as to form a crystallized protective layer 104', in accordance with some embodiments.

[0077] As shown in FIG. 4C, a dielectric layer 306 is deposited over the crystallized protective layer 104' to overfill the trenches, in accordance with some embodiments. The dielectric layer 306 may be made of or include silicon

nitride, silicon oxide, silicon oxynitride, carbon-containing silicon oxide, carbon-containing silicon oxynitride, one or more other suitable materials, or a combination thereof. The dielectric layer 306 may be deposited using an FCVD process. In some embodiments, no seam is formed, as shown in FIG. 4C.

[0078] As shown in FIG. 4D, similar to the process illustrated in FIGS. 3A-3E, the protective caps 206 are formed, and the isolation layer 102 is partially removed, in accordance with some embodiments. As a result, the fin structures 101A to 101C and the dielectric fins 108 protrude from the top surface of the isolation structure (i.e., the remaining portion of the isolation layer 102).

[0079] Afterwards, similar to the embodiments illustrated in FIG. 2A, a dummy gate stack is formed to partially cover the fin structures 101A to 101B and the dielectric fins 108.

[0080] As shown in FIG. 4E, similar to the process illustrated in FIG. 1G, the portions of the fin structures 101A to 101C that are not covered by the dummy gate stack is then partially removed, in accordance with some embodiments. During the recessing of the fin structures 101A to 101C, the inner portion (i.e., the dielectric layer 306) of the dielectric fin 108 under the protective cap 206 is protected and prevented from being damaged by the etchant used for recessing the fin structures 101A to 101C.

[0081] As shown in FIG. 4F, similar to the embodiments illustrated in FIG. 1H, the epitaxial structures 110A to 110C are formed, in accordance with some embodiments. Due to the dielectric fins 108, the epitaxial structures 110A to 110C are prevented from being merged together. The performance and reliability of the semiconductor device structure are ensured.

[0082] Many variations and/or modifications can be made to embodiments of the disclosure. FIG. 5 is a cross-sectional view of an intermediate stage of a process for forming a portion of a semiconductor device structure, in accordance with some embodiments. In some embodiments, no protective cap is formed over the dielectric layer 306.

[0083] In some embodiments, similar to the structure shown in FIG. 1G, the dielectric layer 306 of the dielectric fin 108 is slightly etched during the recessing of the fin structures 101A to 101C. In some embodiments, the topmost surface of the inner portion (i.e., the dielectric layer 306) of the dielectric fin 108 is closer to the semiconductor substrate 100 than the topmost surface of the crystallized protective layer 104' (that functions as a protective shell) of the dielectric fin 108. In some embodiments, the topmost surface of the inner portion (i.e., the dielectric layer 306) has a curved profile. In some embodiments, the topmost surface of the inner portion (i.e., the dielectric layer 306) has convex profile facing downwards, as shown in FIG. 5.

[0084] Embodiments of the disclosure form a semiconductor device structure with dielectric fins. The dielectric fin has a protective shell that extends along sidewalls and a bottom of a dielectric structure. The protective shell may be a crystallized high-k material that has a better etching resistance than the inner portion (i.e., the dielectric structure) of the dielectric fin. The protective shell ensures the structural stability of the dielectric fin. Due to the blocking of the dielectric structure, the seam surrounded by the protective shell is prevented from being merged to form voids through the protective shell. The reliability and performance of the semiconductor device structure are improved.

[0085] In accordance with some embodiments, a semiconductor device structure is provided. The semiconductor device structure includes a first fin structure and a second fin structure over a semiconductor substrate. The semiconductor device structure also includes a first epitaxial structure over the first fin structure and a second epitaxial structure over the second fin structure. The semiconductor device structure further includes a dielectric fin over the semiconductor substrate, and the dielectric fin is between the first fin structure and the second fin structure. The dielectric fin has an inner portion and a protective layer, and the protective layer extends along sidewalls and a bottom of the inner portion. The protective layer has a dielectric constant higher than that of silicon nitride.

[0086] In accordance with some embodiments, a semiconductor device structure is provided. The semiconductor device structure includes an epitaxial structure over a semiconductor substrate. The semiconductor device structure also includes a dielectric fin over the semiconductor substrate. The dielectric fin extends upwards to exceed a bottom surface of the epitaxial structure. The dielectric fin has a dielectric structure and a protective shell, and the protective shell extends along sidewalls and a bottom of the dielectric structure. The protective shell has a first average grain size, and the dielectric structure has a second average grain size. The first average grain size is larger than the second average grain size.

[0087] In accordance with some embodiments, a method for forming a semiconductor device structure is provided. The method includes forming a fin structure over a semiconductor substrate and forming an isolation layer over the fin structure and the semiconductor substrate. The method also includes forming a protective layer over the isolation layer and forming a dielectric layer over the protective layer. The method further includes partially removing the isolation layer, the protective layer, and the dielectric layer. A remaining portion of the isolation layer forms an isolation structure, and remaining portions of the protective layer and the dielectric layer form a dielectric fin. The isolation structure laterally surrounds a lower portion of the fin structure and a lower portion of the dielectric fin. The protective shell has a first average grain size, the dielectric structure has a second average grain size, and the first average grain size is larger than the second average grain size.

[0088] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A semiconductor device structure, comprising:
 - a first semiconductor structure laterally spaced apart from a second semiconductor structure; and
 - a dielectric structure laterally between the first semiconductor structure and the second semiconductor structure, wherein the dielectric structure has an inner por-

tion and an outer layer extending along a sidewall of the inner portion, average grain sizes of the outer layer and the inner portion are different, and a topmost surface of the inner portion is vertically spaced apart from a topmost surface of the outer layer.

2. The semiconductor device structure as claimed in claim 1, wherein the outer layer has a higher dielectric constant than that of silicon nitride.

3. The semiconductor device structure as claimed in claim 1, further comprising:

- a first epitaxial structure over the first semiconductor structure; and
- a second epitaxial structure over the second semiconductor structure.

4. The semiconductor device structure as claimed in claim 3, wherein the first epitaxial structure is in direct contact with the outer layer of the dielectric structure.

5. The semiconductor device structure as claimed in claim 3, wherein the first epitaxial structure and the second epitaxial structure are in direct contact with the outer layer of the dielectric structure.

6. The semiconductor device structure as claimed in claim 1, further comprising:

- a seam extending downwards from a top surface of the inner portion of the dielectric structure.

7. The semiconductor device structure as claimed in claim 6, further comprising a protective cap over the inner portion of the dielectric structure, wherein the protective cap seals the seam, and topmost surfaces of the protective cap and the outer layer are substantially level with each other.

8. The semiconductor device structure as claimed in claim 1, further comprising:

- a protective cap over the inner portion of the dielectric structure, wherein the outer layer extends along a sidewall of the protective cap.

9. The semiconductor device structure as claimed in claim 1, further comprising:

- a gate stack surrounding a portion of the first semiconductor structure, wherein the gate stack extends along a sidewall of the dielectric structure.

10. The semiconductor device structure as claimed in claim 9, further comprising:

- an insulating structure, wherein a portion of the insulating structure extends downwards from a top of the gate stack to reach the dielectric structure.

11. A semiconductor device structure, comprising:

- a semiconductor structure over a substrate;
- a gate stack surrounding the semiconductor structure;
- a dielectric structure beside the gate stack; and
- a protective shell extending along a sidewall of the dielectric structure, wherein average grain sizes of the protective shell and the dielectric structure are different, and topmost surfaces of the dielectric structure and the protective shell are located at different height levels.

12. The semiconductor device structure as claimed in claim 11, wherein the protective shell has a first dielectric constant, the dielectric structure has a second dielectric constant, and the first dielectric constant is higher than the second dielectric constant.

13. The semiconductor device structure as claimed in claim 11, further comprising:

- an insulating layer extending downwards from a top of the gate stack to reach the dielectric structure.

14. The semiconductor device structure as claimed in claim **11**, wherein the protective shell has an outer sidewall and an inner sidewall, the inner sidewall is between the outer sidewall and the dielectric structure, and a lateral distance between the inner sidewall and the outer sidewall is in a range from about 2 nm to about 5 nm.

15. The semiconductor device structure as claimed in claim **11**, further comprising:

a second semiconductor structure laterally spaced apart from the semiconductor structure, wherein a portion of the dielectric structure is between the second semiconductor structure and a portion of the gate stack.

16. The semiconductor device structure as claimed in claim **11**, further comprising:

a protective cap over the dielectric structure, wherein the protective shell covers a sidewall of the protective cap.

17. A semiconductor device structure, comprising:

a first semiconductor structure and a second semiconductor structure over a substrate;

a dielectric structure between the first semiconductor structure and the second semiconductor structure, wherein the dielectric structure has an inner portion and

an outer portion, the outer portion extends along a sidewall of the inner portion, the outer portion has a first average grain size, the inner portion has a second average grain size, and the first average grain size is different than the second average grain size; and

a seam within the dielectric structure.

18. The semiconductor device structure as claimed in claim **17**, further comprising:

a metal gate stack surrounds the first semiconductor structure, wherein a portion of the metal gate stack is between the first semiconductor structure and the dielectric structure.

19. The semiconductor device structure as claimed in claim **18**, further comprising:

an insulating structure extending downwards from a top of the metal gate stack to reach the dielectric structure.

20. The semiconductor device structure as claimed in claim **17**, further comprising:

a protective cap sealing the seam, wherein the outer portion extends along a sidewall of the protective cap.

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